

6 Deformation of solids

AS · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Stress = force/cross-sectional area; strain = extension/original length. Young modulus E = stress/strain in a linear elastic region. The gradient of a force-extension graph is spring constant; its area gives work done. Elastic deformation is reversible on unloading, whereas plastic deformation leaves a permanent extension.

Worked example: follow the reasoning

A wire of length 2.0 m, area $1.0 \times 10^{-6} \text{ m}^2$ extends 1.0 mm under 100 N: stress 10^8 Pa , strain 5×10^{-4} and $E = 2 \times 10^{11} \text{ Pa}$.

Quick check · multiple choice

A Which ratio gives Young modulus?

[1]

A Strain/stress

B Stress/strain

C Force/extension only

D Extension/length only

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 A spring has $k = 80 \text{ N m}^{-1}$ and extends by 0.050 m. Find force and elastic energy stored.

[3]

Final answer: _____

Answers and explanation

Quick check: B. Stress/strain

Question 1 [3 marks]

$F = kx = 4.0 \text{ N}$ [1]. Energy = $\frac{1}{2}kx^2$ [1] = 0.10 J [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Physics 9702, 2025–2027. <https://www.cambridgeinternational.org/Images/664565-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.